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EVEN END SEMESTER EXAMINATION JUNE 2025

Name of the Course: B. Tech.

Semester: II

Name of the Paper: Engineering Mathematics-II

Paper Code: TMA-201

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1 (20 marks)		CO1
a)	Test the convergence of the following series: $\frac{1}{2} + \frac{2}{3}x + \left(\frac{3}{4}\right)^2 x^2 + \left(\frac{4}{5}\right)^3 x^3 + \dots (x > 0)$	
b)	Discuss the nature of the series $1 + \frac{1}{2} \frac{x^2}{4} + \frac{1.3.5}{2.4.6} \frac{x^4}{8} + \frac{1.3.5.7.9}{2.4.6.8.10} \frac{x^6}{12} + \dots (x > 0)$	
c)	Test the convergence of the following series: $\sum_{n=1}^{\infty} (n^4 + 1)^{\frac{1}{4}} - n $	
Q2 (20 marks)		CO2
a)	Solve: $\frac{d^2y}{dx^2} - 4 \frac{dy}{dx} + 4y = 8x^2 e^{2x} \sin 2x$	
b)	Solve: $x^2 \frac{d^2y}{dx^2} + 5x \frac{dy}{dx} + 4y = x^2 + 16 (\log x)^2$	
c)	Solve the given ordinary differential equation using method of variation of parameters: $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + y = (x + 1) e^{2x}$	
Q3 (20 marks)		CO3
a)	Prove that: $\frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$.	
b)	Evaluate the integral $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$.	
c)	Prove that: $\int_{-1}^1 P_m(x) \cdot P_n(x) dx = 0$ if $m \neq n$.	

Q4		(20 marks)	CO 4
a)	Examine the nature of the function $f(z)$, defined by $f(z) = \begin{cases} \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0, \end{cases}$ In the region including the origin.		
b)	If $u - v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function of $z = x + iy$, find the analytic function $f(z)$ in terms of z.		
c)	Find the bilinear transformation that maps $z = 0, -1, i$ onto $w = i, 0, \infty$ respectively. Also find the image of the unit circle $z = 1$.		
Q5		(20 marks)	CO 5
a)	Evaluate : $\oint \frac{z^3 - 6}{2z - 3i} dz$, where the given curve: (i) $z = 2$. (ii) $z = 1$.		
b)	Find the sum of the residues of $f(z) = \frac{\sin z}{z \cos z}$ at its poles inside the circle $z = 2$.		
c)	Integrate the function $\int_0^{2+i} (\bar{z})^2 dz$ along the curve C : (i) Along the line $2y = x$. (ii) From $(0,0)$ to $(2,0)$ and then $(2,0)$ to $(2,1)$.		